Competition

Today's Agenda:

- Quiz
- Predator-Prey Interactions

Interspecific Competition

Occurs when one species limits the growth rate of another through:

1) Reducing resource availability (exploitative/resource competition)

Interspecific Competition

Occurs when one species limits the growth rate of another through:

1) Reducing resource availability (exploitative/resource competition)
2) Reducing access to resources (interference/contest competition)

Interspecific Competition

Occurs when one species limits the growth rate of another through:

1) Reducing resource availability (exploitative/resource competition)
2) Reducing access to resources (interference/contest competition)
3) Increasing shared predators (apparent competition)

Modelling Competition

Lotka-Volterra models again

Modelling Competition

$$\frac{dN_1}{dt} = r_1 N_1 \left(\frac{k_1 - N_1 - \alpha_{12} N_2}{k_1} \right)$$

dN_1/dT = rate of change in N_1
k_1 = carrying capacity of species 1
N_1 = # species 1
N_2 = # species 2
α_{12} = relative impact of sp. 2 on sp. 1

Modelling Competition

$$\frac{dN_1}{dt} = r_1N_1\left(\frac{k_1 - N_1 - \alpha_{12}N_2}{k_1}\right)$$

$$\frac{dN_2}{dt} = r_2N_2\left(\frac{k_2 - N_2 - \alpha_{21}N_1}{k_2}\right)$$

dN_1/dT = rate of change in N_1
k_1 = carrying capacity of species 1
N_1 = # species 1
N_2 = # species 2
α_{12} = relative impact of sp. 2 on sp. 1

dN_2/dT = rate of change in N_2
k_2 = carrying capacity of species 2
α_{21} = relative impact of sp. 1 on sp. 2

Modelling Competition

α_{ij} = competition coefficient

$$\frac{dN_1}{dt} = r_1 N_1 \left(\frac{k_1 - N_1 - \alpha_{12} N_2}{k_1} \right)$$

$$\frac{dN_2}{dt} = r_2 N_2 \left(\frac{k_2 - N_2 - \alpha_{21} N_1}{k_2} \right)$$

dN_1/dT = rate of change in N_1
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α_{12} = relative impact of sp. 2 on sp. 1

dN_2/dT = rate of change in N_2
k_2 = carrying capacity of species 2
α_{21} = relative impact of sp. 1 on sp. 2

Zero Net Growth Isoclines (ZNGIs)

- We can ask under what conditions populations are stable
- These are isoclines of zero net growth
- Set net growth to zero and solve for N1

Zero Net Growth Isoclines (ZNGIs)

$$\frac{dN_1}{dt} = r_1N_1\left(\frac{k_1 - N_1 - \alpha_{12}N_2}{k_1}\right)$$

$$\frac{dN_2}{dt} = r_2N_2\left(\frac{k_2 - N_2 - \alpha_{21}N_1}{k_2}\right)$$

Zero Net Growth Isoclines (ZNGIs)

$$0 = r_1 N_1 \left(\frac{k_1 - N_1 - \alpha_{12} N_2}{k_1} \right)$$

$$0 = r_2 N_2 \left(\frac{k_2 - N_2 - \alpha_{21} N_1}{k_2} \right)$$

Zero Net Growth Isoclines (ZNGIs)

$$0 = \left(\frac{k_1 - N_1 - \alpha_{12} N_2}{k_1} \right)$$

$$0 = \left(\frac{k_2 - N_2 - \alpha_{21} N_1}{k_2} \right)$$

Zero Net Growth Isoclines (ZNGIs)

$$0 = k_1 - N_1 - \alpha_{12}N_2$$

$$0 = k_2 - N_2 - \alpha_{21}N_1$$

Zero Net Growth Isoclines (ZNGIs)

$N_1 = k_1 - \alpha_{12}N_2$

$\alpha_{21}N_1 = k_2 - N_2$

Zero Net Growth Isoclines (ZNGIs)

$$N_1 = k_1 - \alpha_{12}N_2$$

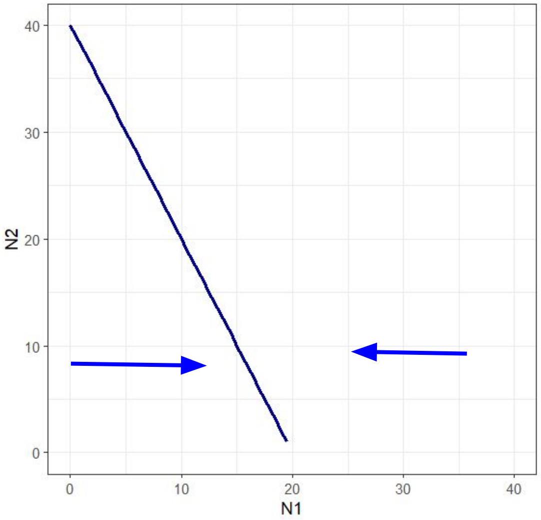
$$N_1 = \frac{k_2 - N_2}{\alpha_{21}}$$

Zero Net Growth Isoclines (ZNGIs)

Species 1

$$N_1 = k_1 - \alpha_{12}N_2$$

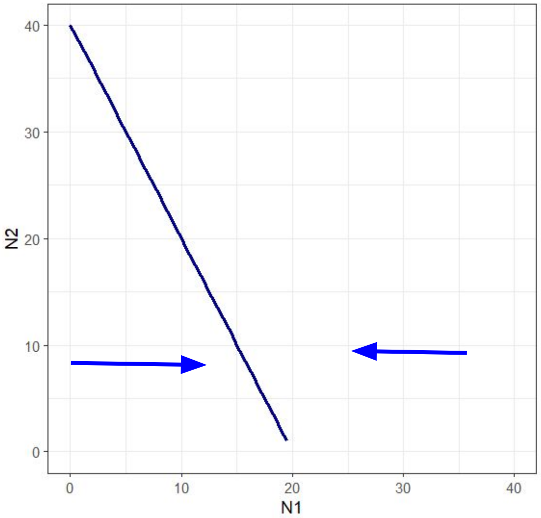
$$N_1 = \frac{k_2 - N_2}{\alpha_{21}}$$



Zero Net Growth Isoclines (ZNGIs)

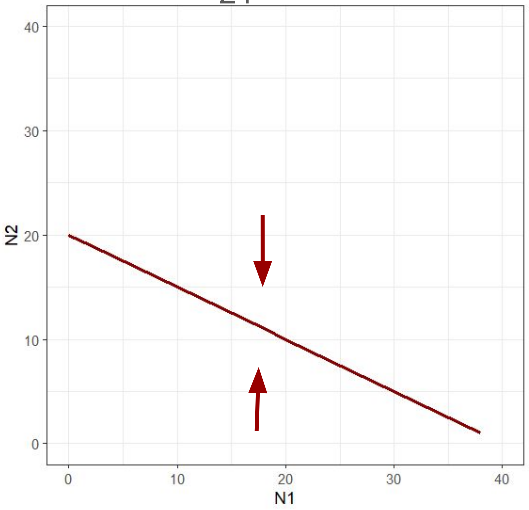
Species 1

$$N_1 = k_1 - \alpha_{12}N_2$$



Species 2

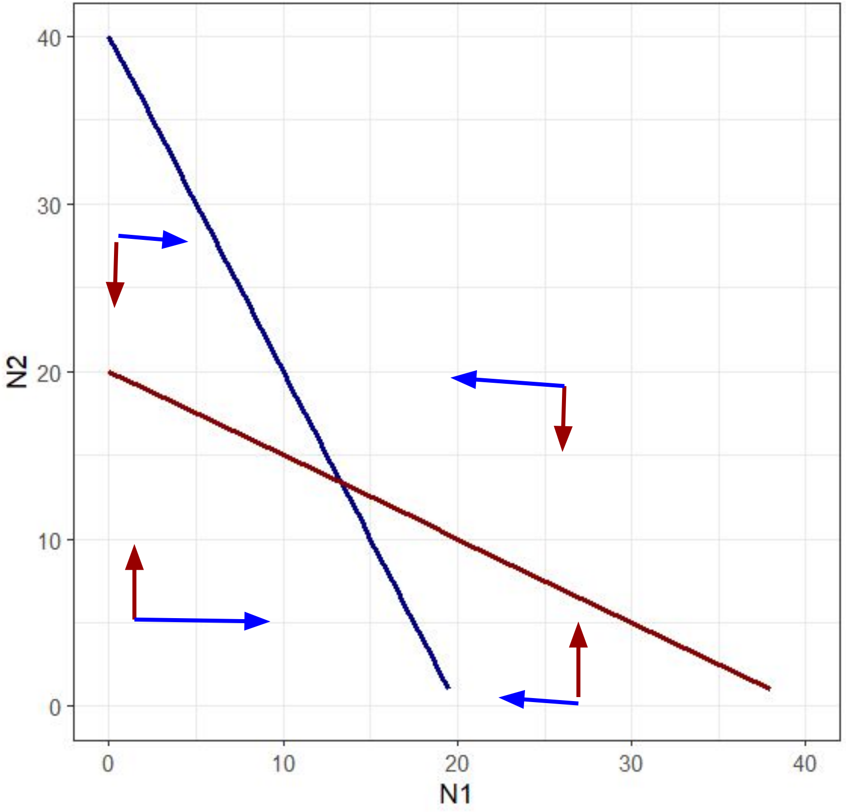
$$N_1 = \frac{k_2 - N_2}{\alpha_{21}}$$



Zero Net Growth Isoclines (ZNGIs)

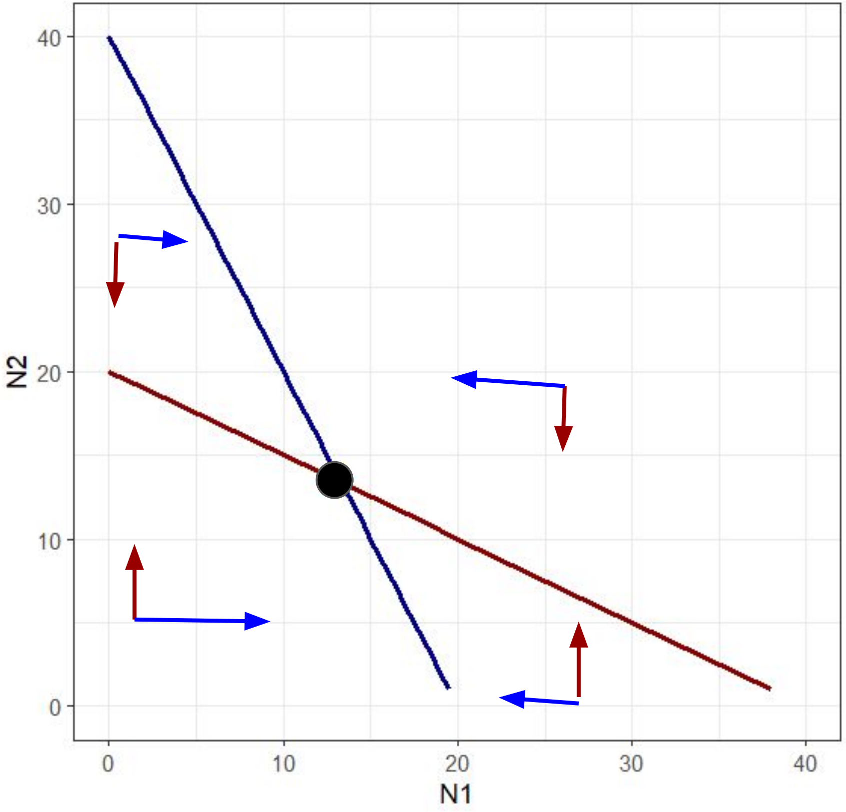
Species 1

Species 2



Zero Net Growth Isoclines (ZNGIs)

Species 1 **Species 2**

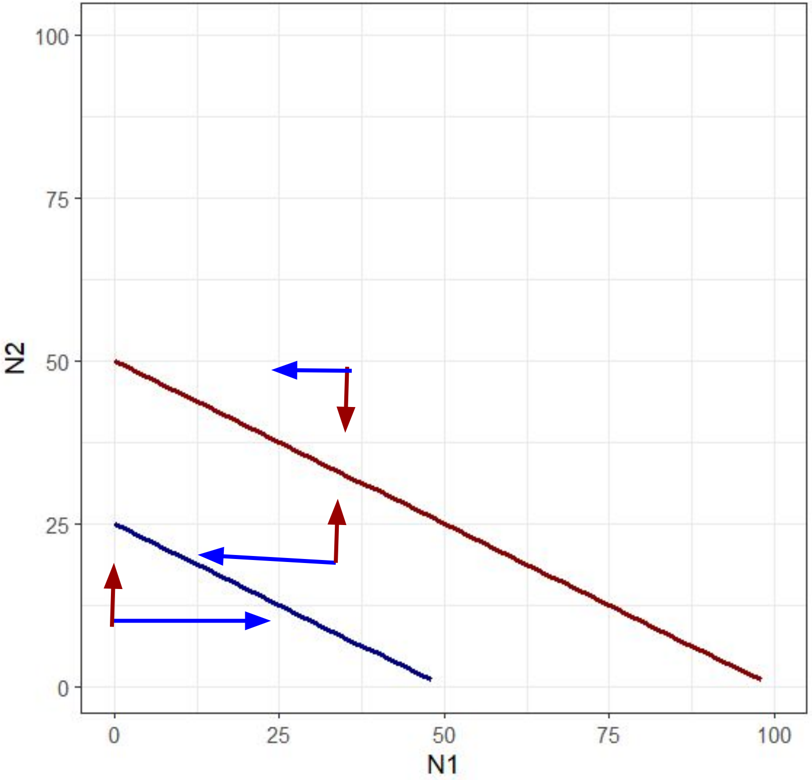


Stable coexistence

- Both species can invade from low density

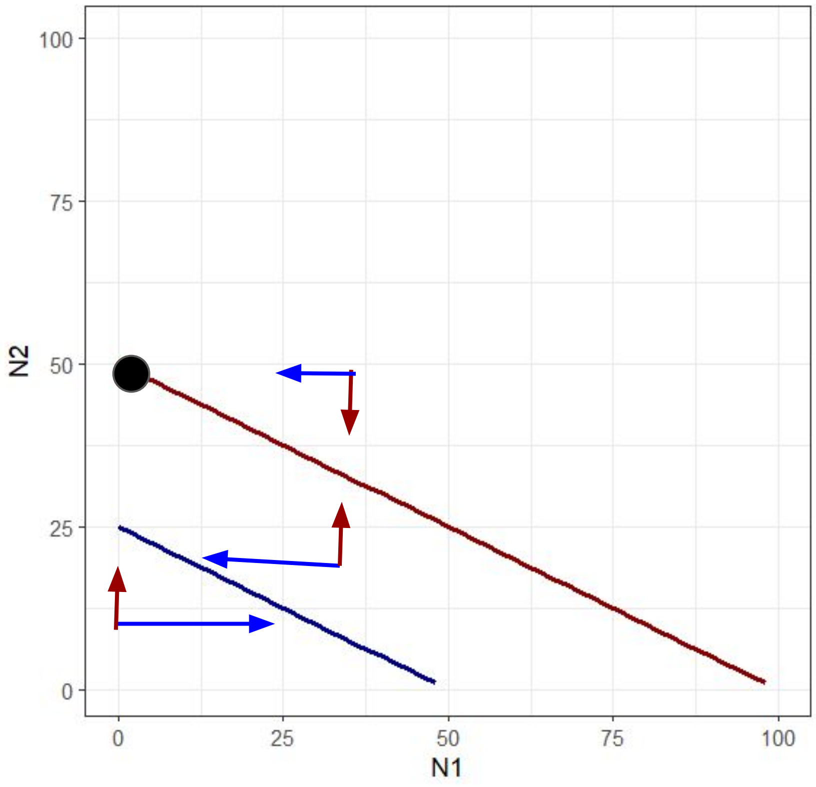
Zero Net Growth Isoclines (ZNGIs)

Species 1 **Species 2**



Zero Net Growth Isoclines (ZNGIs)

Species 1 **Species 2**

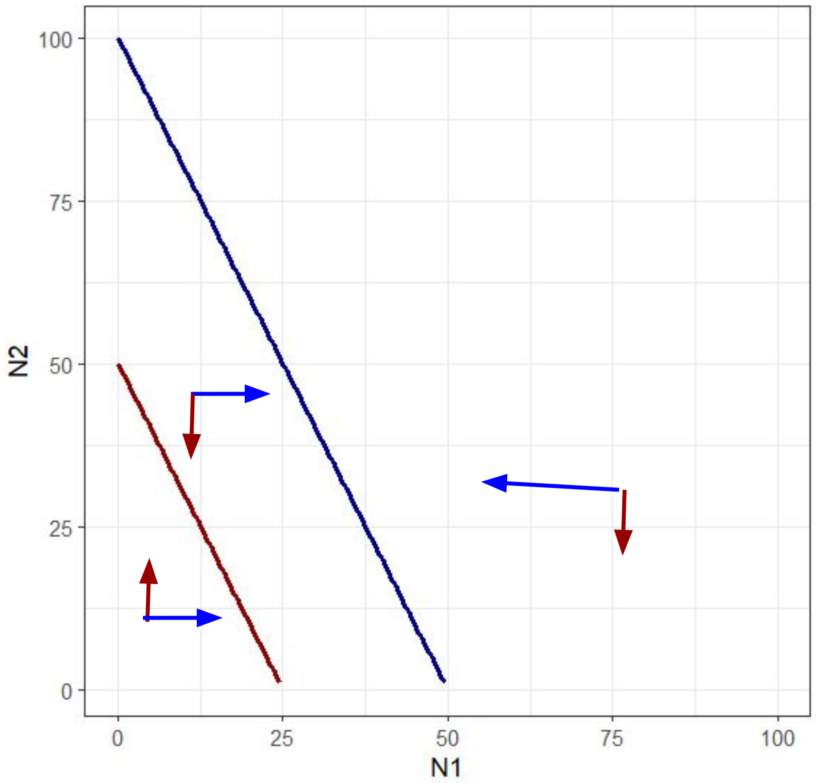


Species 2 excludes species 1

Zero Net Growth Isoclines (ZNGIs)

Species 1

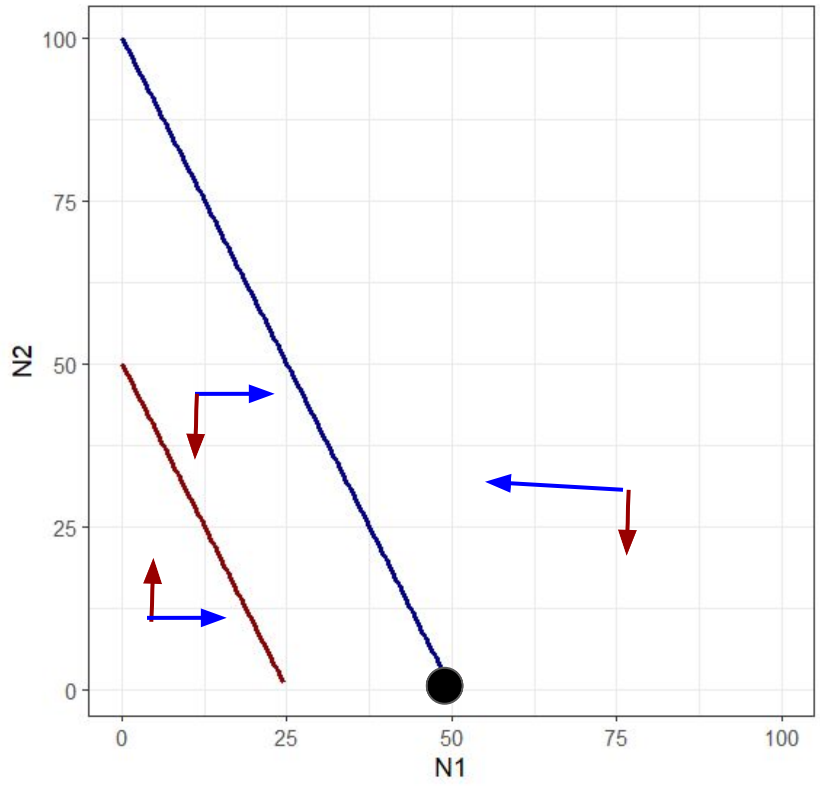
Species 2



Zero Net Growth Isoclines (ZNGIs)

Species 1

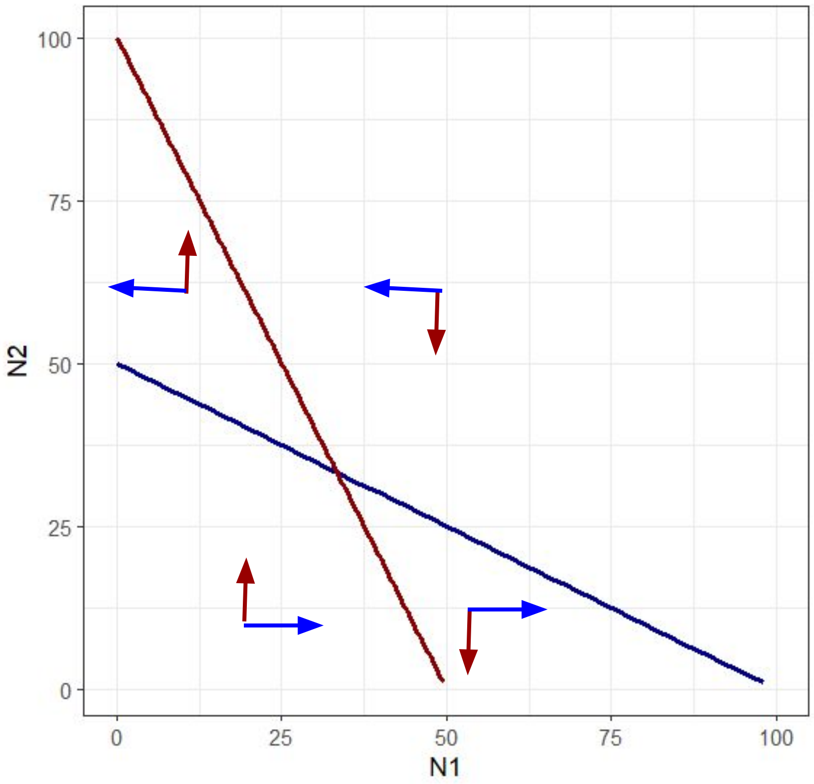
Species 2



Species 1 excludes species 2

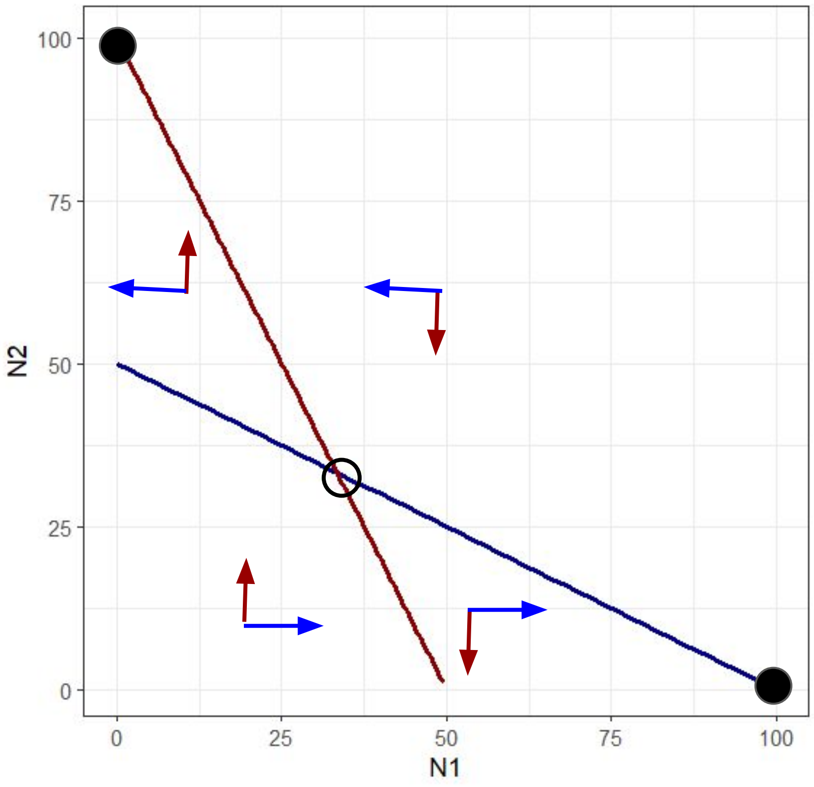
Zero Net Growth Isoclines (ZNGIs)

Species 1 **Species 2**



Zero Net Growth Isoclines (ZNGIs)

Species 1 **Species 2**



Unstable coexistence!

- If nudged away from equilibrium, one species or the other can be excluded!

ZNGIs can be confusing

Chesson 2000 provides are more intuitive formulation

$$\frac{1}{N_i} \cdot \frac{dN_i}{dt} = r_i\,(1 - \alpha_{ii}N_i - \alpha_{ij}N_j)$$

Annu. Rev. Ecol. Syst. 2000. 31:343–66

MECHANISMS OF MAINTENANCE OF SPECIES DIVERSITY

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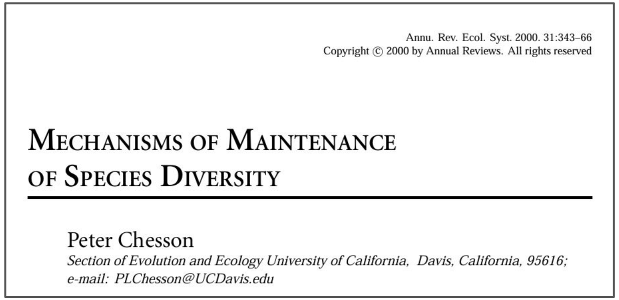
Chesson 2000 provides are more intuitive formulation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$$\frac{1}{N_i} \cdot \frac{dN_i}{dt} = r_i\,(1 - \alpha_{ii}N_i - \alpha_{ij}N_j)$$

α’s here are absolute impact.
α_{ii} = intraspecific competition
α_{ij} = interspecific competition

Coexistence criteria

Both species need to be able to invade when rare, when other species is present

$$\frac{1}{N_1} \cdot \frac{dN_1}{dt} = r_1 (1 - \alpha_{11}N_1 - \alpha_{12}N_2)$$

α's here are absolute impact.
α_{ii} = intraspecific competition
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Coexistence criteria

Both species need to be able to invade when rare, when other species is present

$$0 < r_1 (1 - \alpha_{11} 0 - \alpha_{12} N_2)$$

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Coexistence criteria

Both species need to be able to invade when rare, when other species is present

$0 \quad < \quad 1 - \alpha_{12}N_2$

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Coexistence criteria

Both species need to be able to invade when rare, when other species is present

$$0 < 1 - \alpha_{12}N_2$$

$$\frac{1}{N_2} \cdot \frac{dN_2}{dt} = r_2 (1 - \alpha_{22}N_2 - \alpha_{21}N_1)$$

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Coexistence criteria

Both species need to be able to invade when rare, when other species is present

$$0 < 1 - \alpha_{12}N_2$$

$$0 = r_2 (1 - \alpha_{22}N_2 - \alpha_{21}0)$$

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Coexistence criteria

Both species need to be able to invade when rare, when other species is present

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Coexistence criteria

Both species need to be able to invade when rare, when other species is present

$0 \quad < \quad 1 - \alpha_{12}N_2$

$\alpha_{22}N_2 = \quad 1$

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Coexistence criteria

Both species need to be able to invade when rare, when other species is present

$$0 < 1 - \alpha_{12} N_2$$

$$N_2 = \frac{1}{\alpha_{22}}$$

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Both species need to be able to invade when rare, when other species is present

$\alpha_{12}N_2 < 1$

$$N_2 = \frac{1}{\alpha_{22}}$$

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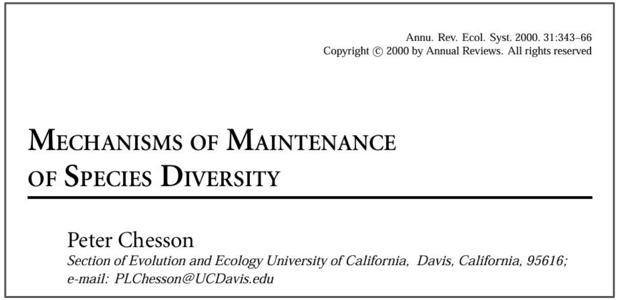
In order to coexist:

$\alpha_{ii} > \alpha_{ji}$

$\alpha_{jj} > \alpha_{ij}$

α_{ii} = intraspecific competition
α_{ij} = interspecific competition

Coexistence criteria



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In order to coexist:

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In other words: **to coexist, species need to limit themselves more than they limit their competitors**

α_{ii} = intraspecific competition
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In order to coexist:

$\alpha_{ii} > \alpha_{ji}$

$\alpha_{jj} > \alpha_{ij}$

IMPORTANT:
WILL BE ON A TEST

In other words: **to coexist, species need to limit themselves more than they limit their competitors**

α_{ii} = intraspecific competition
α_{ij} = interspecific competition

Resource competition

So far we've been talking about competition in terms of reduction in growth rate

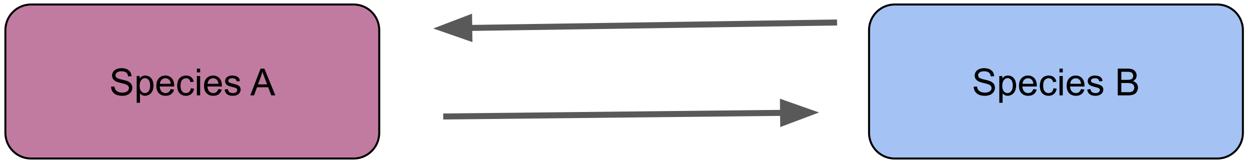
- But competition usually isn't actually species interacting with each other
- It usually involves shared resources
- Resource models are more mechanistic

Resource competition

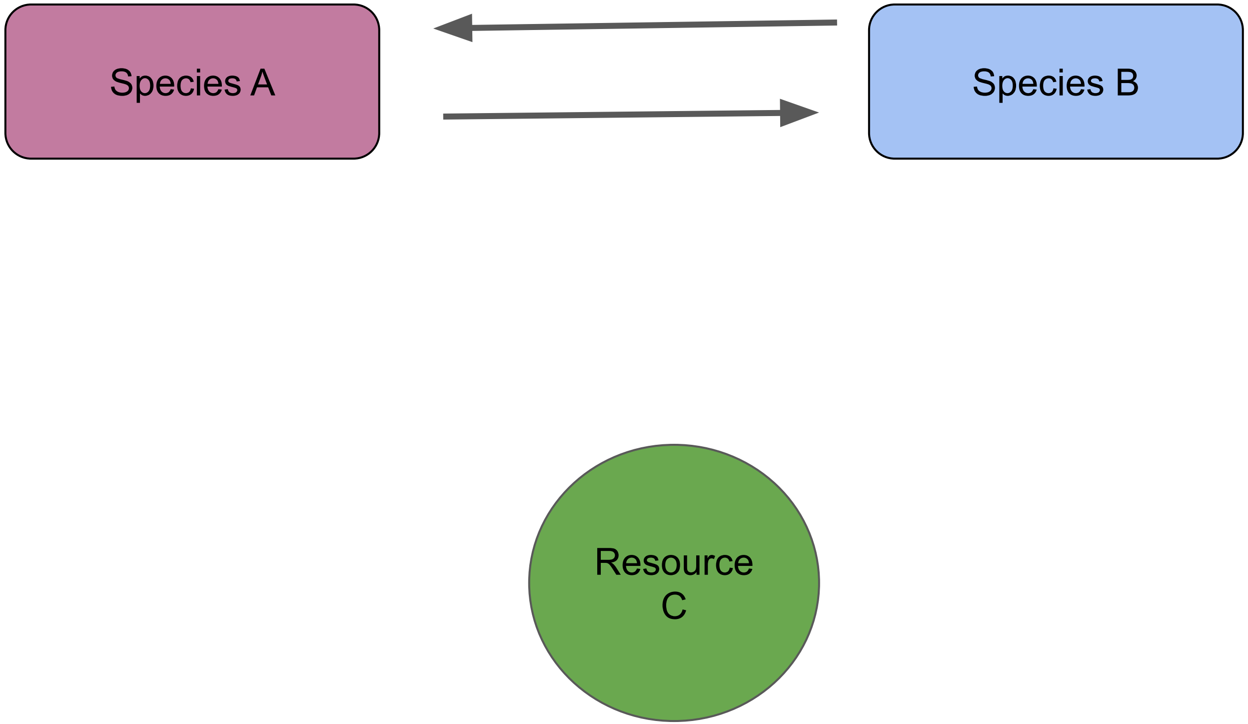
Species A

Species B

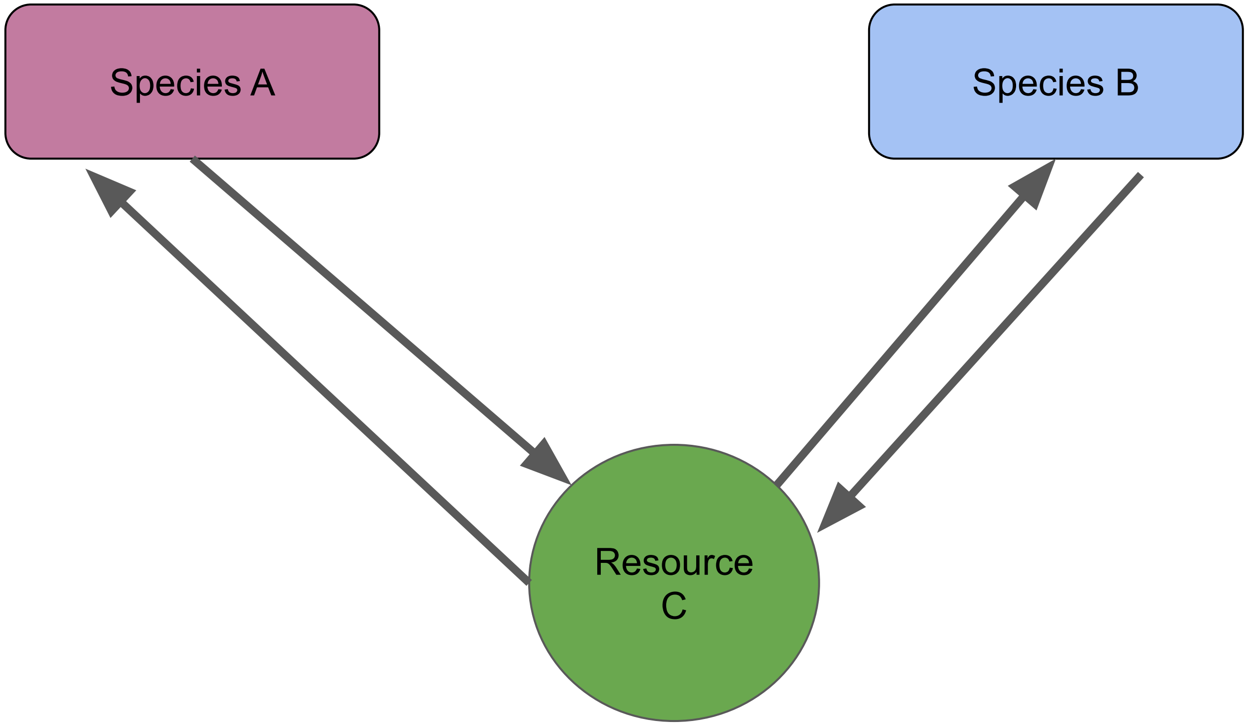
Resource competition



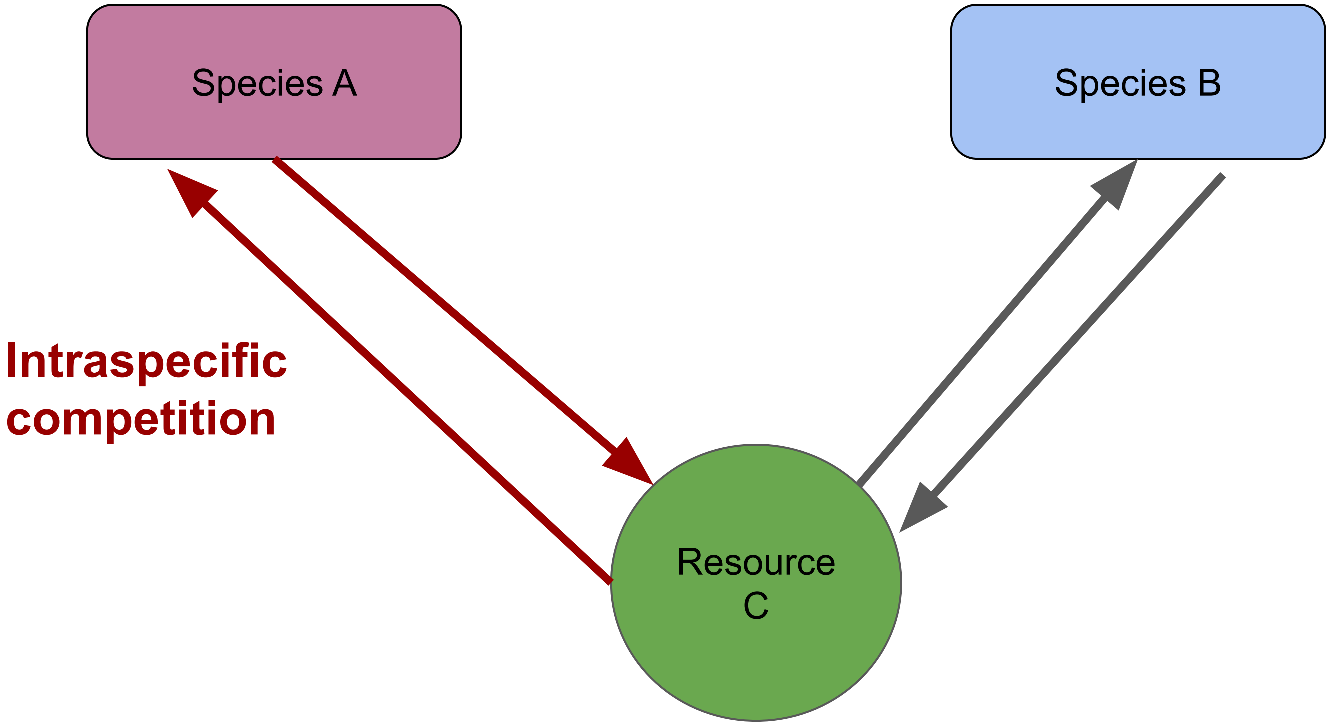
Resource competition



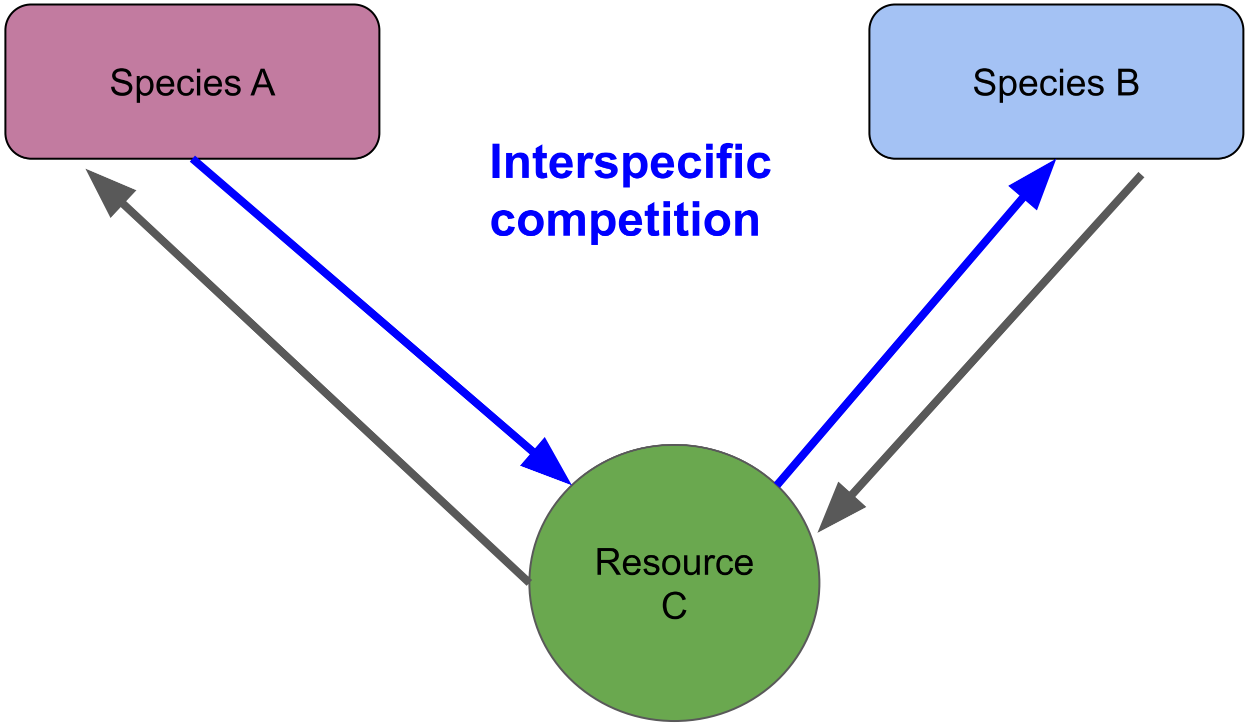
Resource competition



Resource competition



Resource competition



Resource competition models

What is a resource?

Resource competition models

What is a resource?

- Increases population growth
- Is consumed by use

Resource competition models

What is a resource?

- Increases population growth
- Is consumed by use

Resources:
Food
Nutrients
Water (but trivial in aquatic)
Light (for producers)
Oxygen (but trivial in terrestrial)

Resource competition models

What is a resource?

- Increases population growth
- Is consumed by use

Resources:
Food
Nutrients
Water (but trivial in aquatic)
Light (for producers)
Oxygen (but trivial in terrestrial)

NOT Resources:
Temperature
Elevation
Pressure

R* Theory

$$\frac{1}{N_1} \cdot \frac{dN_i}{dt} = a_i R - d_i$$

R = resource availability
a_i = resource consumption rate of species i
d_i = death rate of species i

R* Theory

What is the minimum value of R needed?

$$\frac{1}{N_1} \cdot \frac{dN_i}{dt} = a_i R - d_i$$

R = resource availability
a_i = resource consumption rate of species i
d_i = death rate of species i

R* Theory

What is the minimum value of R needed?

$$0 = a_i R - d_i$$

R = resource availability
a_i = resource consumption rate of species i
d_i = death rate of species i

R* Theory

What is the minimum value of R needed?

$$\frac{d_i}{a_i} = R$$

R = resource availability
a_i = resource consumption rate of species i
d_i = death rate of species i

R* Theory

What is the minimum value of R needed?

$$R^* = \frac{d_i}{a_i}$$

R = resource availability
a_i = resource consumption rate of species i
d_i = death rate of species i

R* Theory

$$R^* = \frac{d_i}{a_i}$$

- Minimum resources required
- Will determine equilibrium abundance
    - Population will increase until it draws resources down to this level
- Determined by death rate, consumption rate

R = resource availability
a_i = resource consumption rate of species i
d_i = death rate of species i

R* determines who wins in competition

$$R^* = \frac{d_i}{a_i}$$

The species with the lowest R* wins

- Will draw resource down to a level that excludes all other species

R = resource availability
a_i = resource consumption rate of species i
d_i = death rate of species i

R* determines who wins in competition

$$R^* = \frac{d_i}{a_i}$$

The species with the lowest R* wins

- Will draw resource down to a level that excludes all other species
- Note this is only true for species limited by a single resource under constant conditions
- But is still useful generally

R = resource availability
a_i = resource consumption rate of species i
d_i = death rate of species i

Competition for multiple resources

- Gets a lot trickier
- Resources can be essential vs substitutable

Competition for multiple resources

- In order to stably coexist
  - Species need to differ in resource usage
  - Species need to have a larger impact on the resources that are most limiting for them

Competition for multiple resources

- In order to stably coexist
  - Species need to differ in resource usage
  - Species need to have a larger impact on the resources that are most limiting for them

These conditions lead to **species limiting themselves more than they limit their competitors** ($\alpha_{ii} > \alpha_{ji}$)

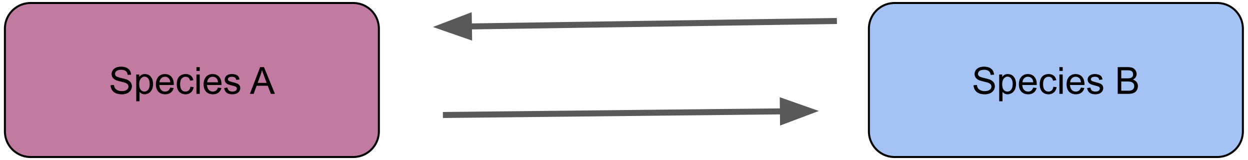
Competition for multiple resources

- In order to stably coexist
  - Species need to differ in resource usage
  - Species need to have a larger impact on the resources that are most limiting for them

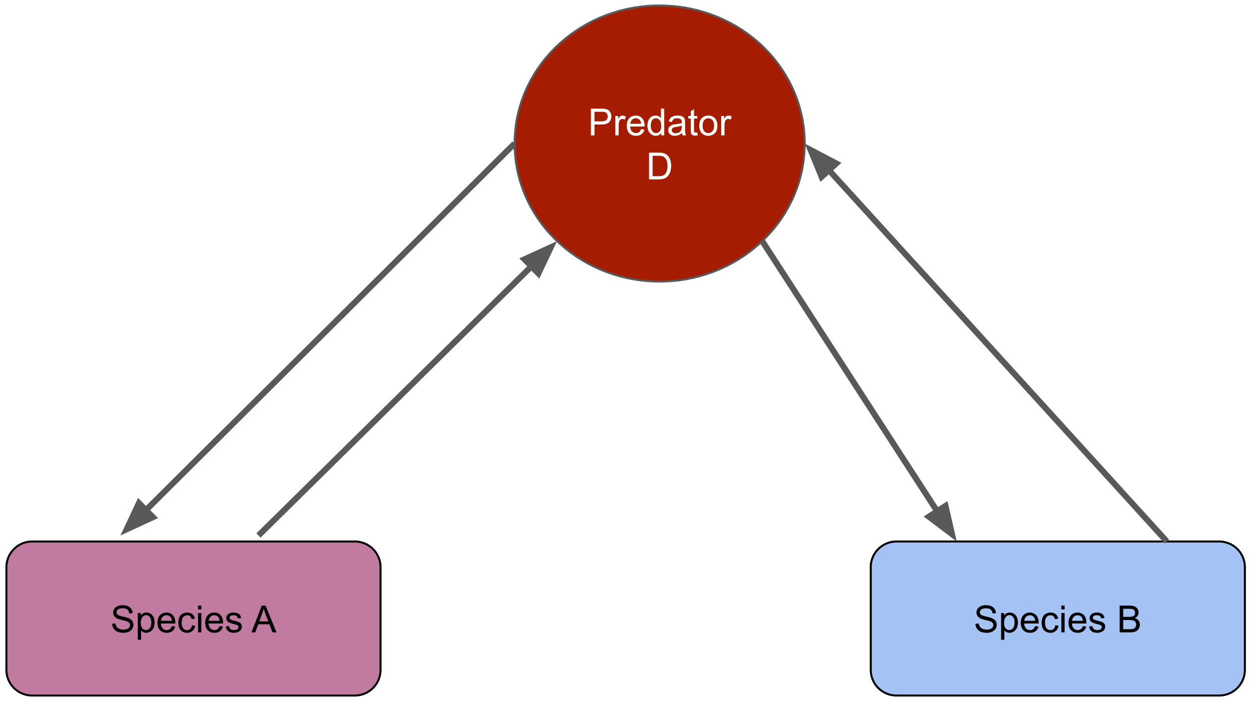
These conditions lead to **species limiting themselves more than they limit their competitors** ($\alpha_{ii} > \alpha_{ji}$)

When coexistence is purely driven by resource differences, the number of species that can be supported is equal to the number of different resources

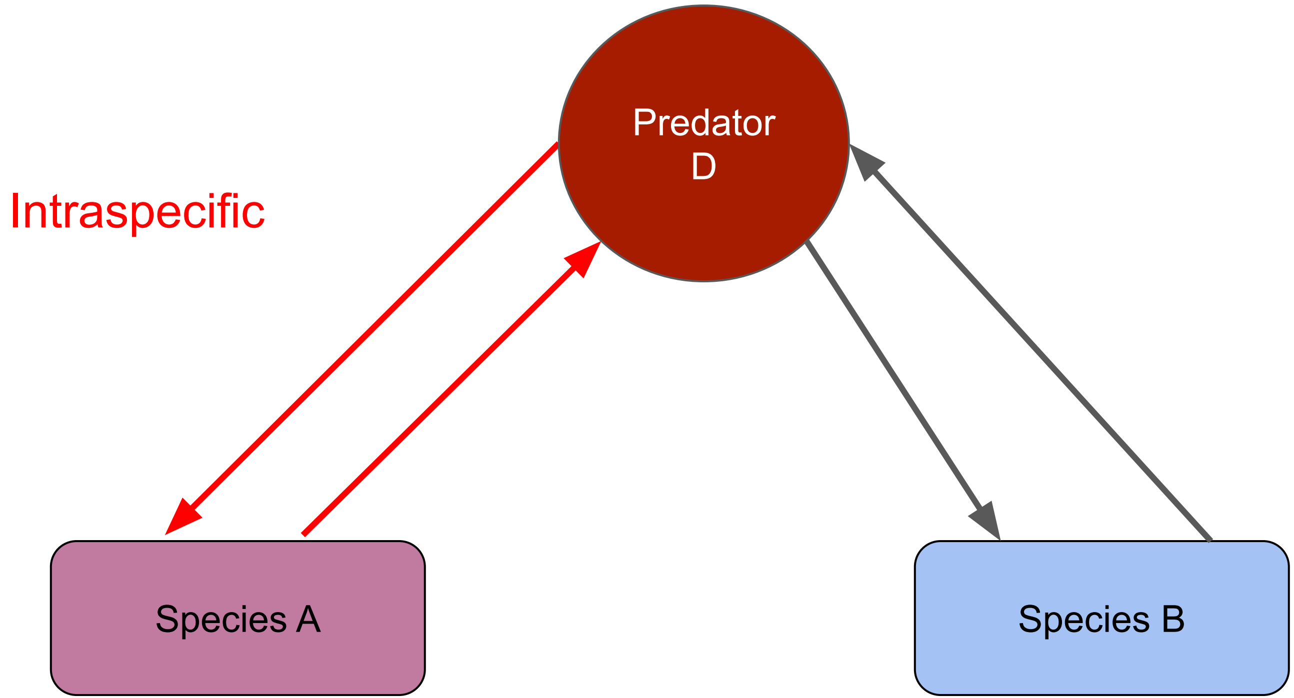
Apparent Competition



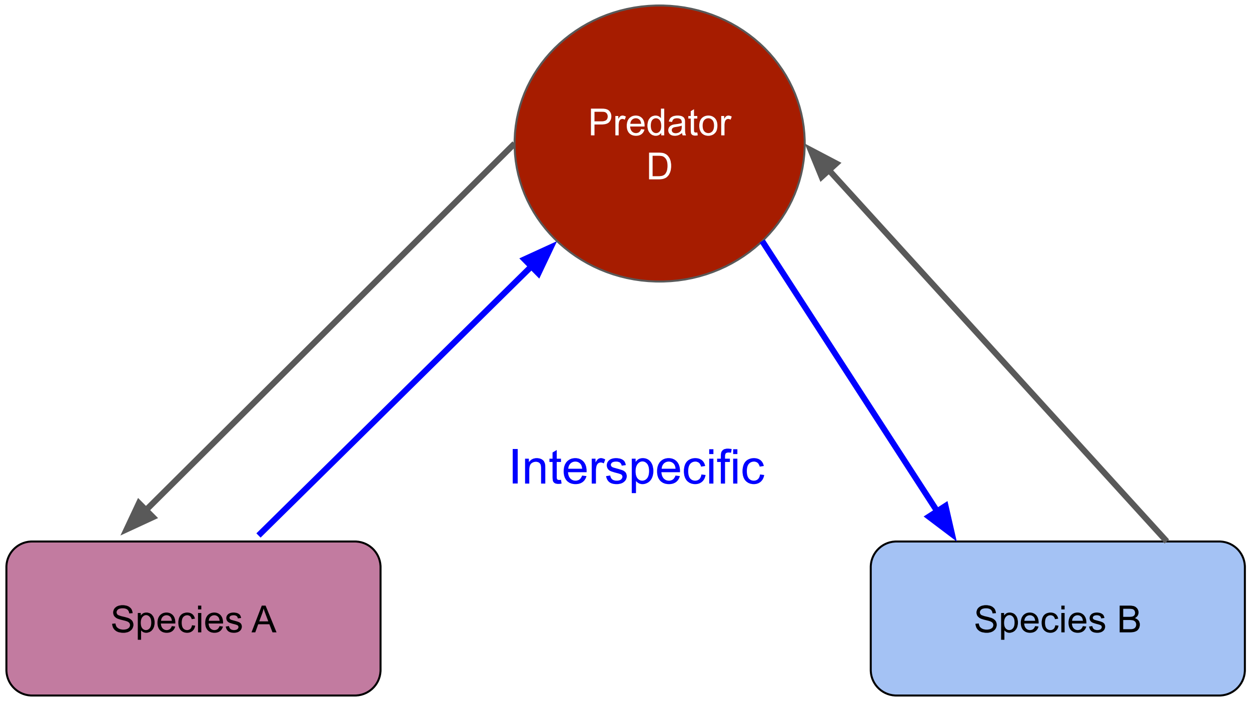
Apparent Competition



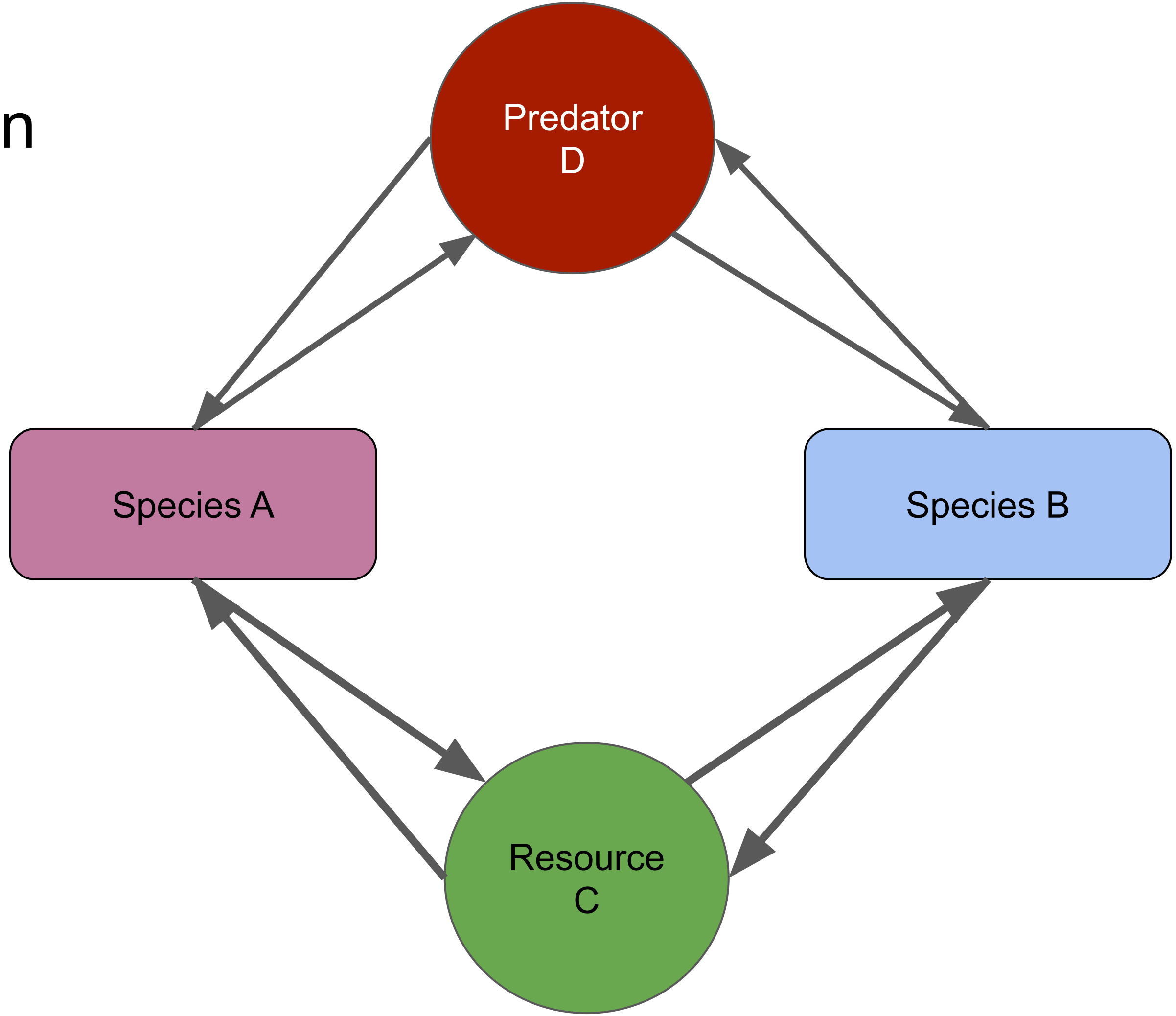
Apparent Competition

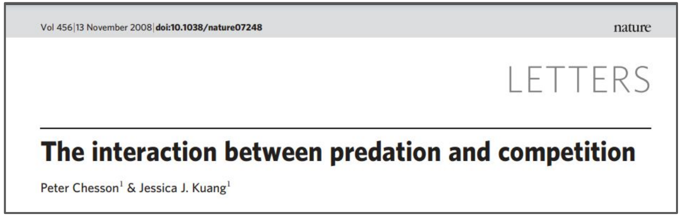


Apparent Competition



Apparent Competition




Vol 456 | 13 November 2008 | doi:10.1038/nature07248
nature

LETTERS

The interaction between predation and competition

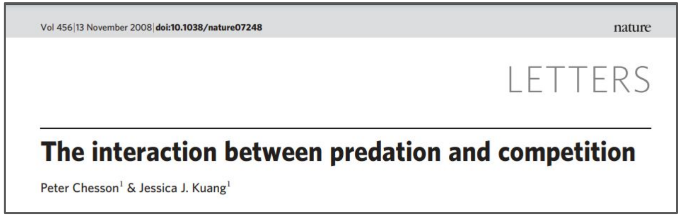
Peter Chesson[1] & Jessica J. Kuang[1]

Apparent Competition

Competition and Predation are symmetrical!

Promote coexistence in the same ways!

2 species can coexist with
2 predators, 2 resources,
1 predator and 1 resource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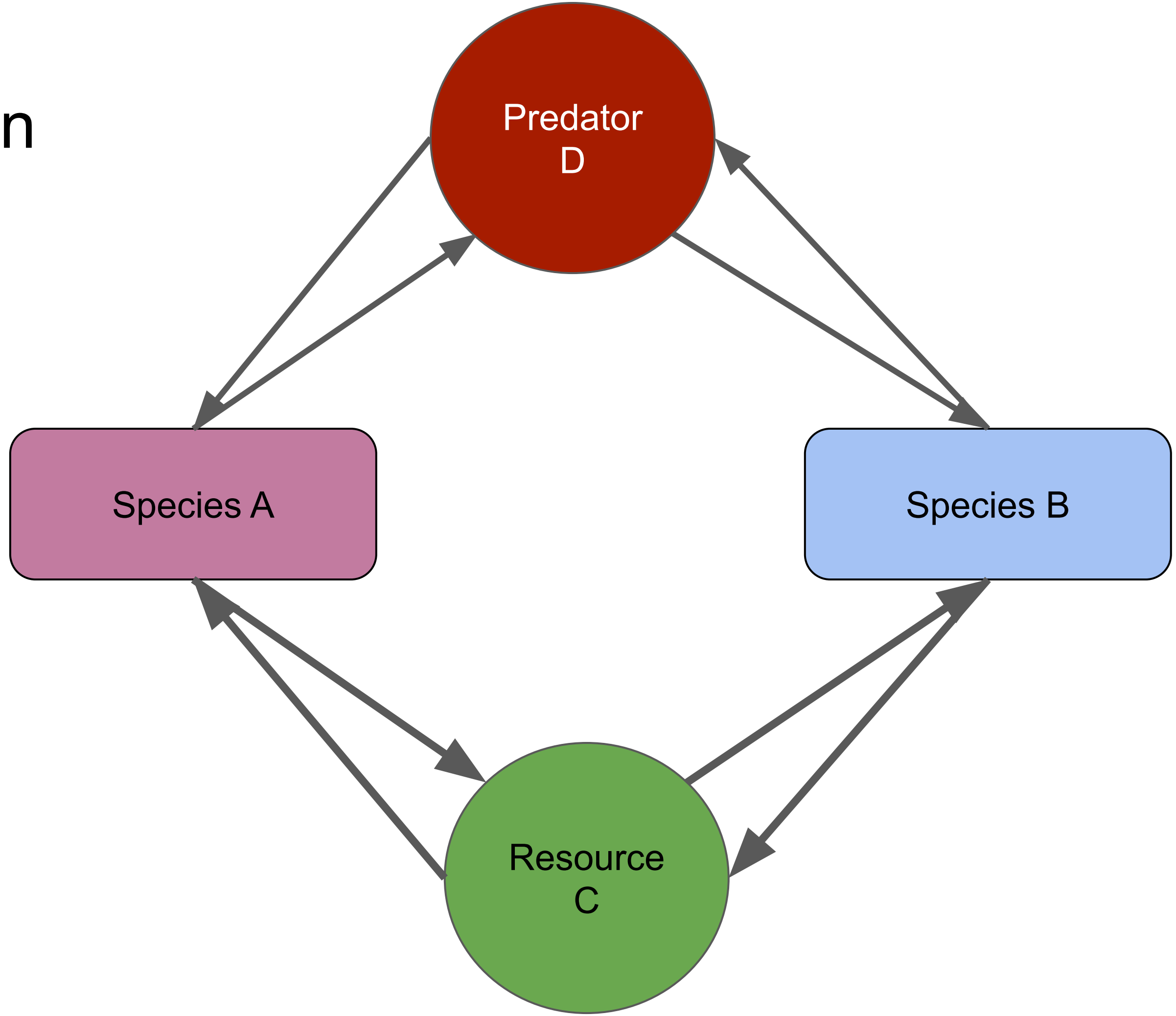

Main Takeaways

- To coexist, species need to limit their own growth more than their competitors
- Species that can exist on the lowest amount of a resource will win
- Predation is symmetrical with competition

Next class: Coexistence